AP CALCULUS AB

Unit 1 • Self Check 1.3 • Lessons 1-5 and 1-6 • Answer key

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Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

f is continuous, $f(0) = 3$, $f(4) = -1$. Which must be true?

(A) $f(2) = 1$
(B) $f(c) = 0$ for some c in $(0, 4)$
(C) exactly one zero
(D) $f(c) = 4$

Solution

Answer: (B).

Question 2

NO CALCULATOR

Why does $\frac{1}{x}$ on $[-1, 1]$ never equal 0, although $f(-1) < 0 < f(1)$?

(A) 0 is not between them
(B) IVT is only for polynomials
(C) not continuous on $[-1, 1]$
(D) it is decreasing

Solution

Answer: (C).

Question 3

NO CALCULATOR

What is the average rate of change of $h(x) = \frac{12}{x}$ on $[2, 6]$?

(A) -1
(B) -4
(C) 1
(D) $-\frac{1}{2}$

Solution

Answer: (A).

Question 4 (Free response) NO CALCULATOR

Find the absolute maximum and minimum of $f(x) = x^2 - 4x + 5$ on $[0, 3]$, and explain why they must exist. Then find the average rate of change of f on $[0, 3]$.

Solution
Continuous on a closed interval (EVT). Max 5 at $x = 0$, min 1 at $x = 2$. Average rate $\frac{2-5}{3} = -1$.